

Problem Set 5

Instructor: Robert Rand

1 Divisibility

Problem 1: The following common properties of a relation R are defined as follows

1. *reflexivity*: $\forall x, Rxx$
2. *symmetry*: $\forall x y, Rxy \rightarrow Ryx$
3. *asymmetry*: $\forall x y, Rxy \rightarrow \neg Ryx$
4. *antisymmetry*: $\forall x y, (Rxy \wedge Ryx) \rightarrow x = y$
5. *transitivity*: $\forall x y, (Rxy \wedge Ryz) \rightarrow Rxz$

For each property, state whether it holds of the *divides* relation restricted to natural numbers (not integers). Prove it (a counterexample suffices where false).

2 Euclid's Algorithm

Problem 2: For each of the following, use Euclid's algorithm to calculate the greatest common divisor. Show each step.

1. $GCD(15, 18)$
2. $GCD(29, 16)$
3. $GCD(120, 105)$

3 Modular Arithmetic

Problem 3: Define the properties “even” and “odd” in terms of congruences modulo m ($\cong_m$).

Problem 4: Suppose we defined $a <_m b$ as $\exists x > 0, a + x \cong_m b$. Show that this would imply $\forall ab, a <_m b$.